

Distribution system planning to accommodate distributed energy resources and PEVs



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ABSTRACT

With deregulation of the power industry, environmental policy changes, advancements in technology, and the transformation to smart grid, the distribution planning paradigm has gone through significant changes in recent years. Concurrently, with increase in gas prices, driven by a foreseeable fossil fuel depletion in the future, developments in the automotive sector, and environmental concerns, penetration of plug-in electric vehicles (PEVs) has been increasing. These changes will continue to drive the distribution planning problem to evolve in the coming years. This paper presents a comprehensive long-term distribution planning framework from the perspective of local distribution companies (LDCs) considering distributed generation (DG), substations, capacitors, and feeders. Apart from considering the usual demand profile, the proposed framework considers uncontrolled and controlled (smart) PEV charging demand, as well as demand response (DR) options. Based on a back-propagation algorithm combined with cost-benefit analysis, a novel approach is proposed to determine the optimal upgrade plan, allocation, and sizing of the selected components in distribution systems, to minimize the total capital and operating cost. A new iterative method is proposed which involves post-processing the plan decisions to guarantee acceptable adequacy levels for each year of the planning horizon. The performance of the proposed framework is examined considering several case studies on the 33-bus and 69-bus test systems. It is noted from the studies that the presence of uncontrolled PEV charging loads results in much higher plan costs as compared to the case without PEVs, impacts the distribution plan significantly, and hence these loads indeed need be considered in the planning process. On the other hand, smart charging of PEVs have a much reduced impact on the plan cost, and helps alleviate some investments which were needed with uncontrolled charging PEVs. Therefore, the LDCs and policy makers need to encourage customers to adapt these options in the long-run.

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1. Introduction

Over the last decade, a major change in paradigm has taken place in power system planning because of deregulation of the power industry, policy changes driven by concerns on environmental emissions, advancement in technology including renewable energy generation, the use of two-way communication between customers and utility, and the consequent transformation of the grid to a self-healing and intelligent system, referred to as a smart grid. These changes have a significant bearing on the electrical distribution systems with increasing penetration of demand

response (DR), distributed generation (DG), and energy storage systems (ESS), together, known as distributed energy resources (DERs), and proposed as part of the solution to today's energy and environmental challenges. Since DERs are mainly targeted at the distribution system level, design and planning issues at this level are key features for the development of the future distribution system [1].

With the increase in gas prices and environmental concerns, penetration of plug-in electric vehicles (PEVs) has been increasing in recent times. Therefore, it is important to understand how distribution planning is impacted when uncontrolled PEV loads penetrate into the distribution system, or how PEV smart charging can help reduce the plan burden of the local distribution companies (LDCs). It is also necessary to examine how other DERs penetrate into the distribution system vis-a-vis PEV penetration, over the long-term, and the distribution plan adjusted to suit such

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Nomenclature

Indices

b	index for load block, $b = 1, 2, \dots, B$
i, j	index for buses, $i = 1, 2, \dots, N$
N	total number of system buses
SS	subset of substation buses ($SS \in i$)
t	index for year, $t = 1, 2, \dots, T$

Parameters

BL	Budget limit, \$
C^C	capital cost of capacitor, \$/p.u.
$C^{DG.F}$	capital cost of DG unit, \$/p.u.
$C^{DG.O}$	operating cost of DG units, \$/p.u.
C^{DR}	payment towards responsive demand, \$/p.u.
$C^{Fdr.F}, C^{Fdr.V}$	fixed and variable component of capital cost of feeder, \$ and \$/p.u.
$C^{SS.F}, C^{SS.V}$	fixed and variable component of capital cost of substation, \$ and \$/p.u.
C^{UN}	cost of unserved energy, \$/p.u.
CBL	capacitor budget limit, \$
DG^{CapMax}	maximum allowable DG capacity, p.u.
DG^{NMax}	maximum number of installed DG units
E	energy needed to charge a PEV, p.u.
$Ge_{i,j}$	geographic cost factor of feeder i – j , \$
Hr_b	hours per day in load block b
$Le_{i,j}$	length of feeder between i and j , km
M	big number used in MIP modelling
NH_i	total number of houses at bus i
p^{PEVCap}	maximum allowable power drawn by PEV during charging, p.u.
$p^{Fdr}_{i,j,b}, Q^{Fdr}_{i,j,b}$	active and reactive power flow from i to j , p.u.
$p^{PEVUNC}_{i,b}$	power drawn by PEV in uncontrolled mode, p.u.
$Pd_{i,b}, Qd_{i,b}$	active and reactive power demand, p.u.
PEV^{pen}	the percentage of PEV penetration
Q^{CMax}	maximum capacitor size, p.u.
$S^{Fdr}_{i,j}$	existing feeder capacity, p.u.
S^{SSCap}, P^{SSCap}	existing substation capacity, p.u.
V^{Min}, V^{Max}	minimum and maximum allowable voltage, p.u.
$Y_{i,j}$	magnitude of admittance matrix element, p.u.
γ	capacity reserve margin with respect to Base Case peak load
$\theta_{i,j}$	angle of bus admittance matrix element, rad
ρ_b	electricity market price, \$/ p.u.
α_0	share of demand that is available for DR, p.u.

Variables

$p^{DG}_{i,b}$	capacity of DG unit, p.u.
$p^{DG}_{i,b}$	power generated from DG unit, p.u.
$p^{DR}_{i,b}$	power contributed by DR participants, p.u.
$p^{Fdr}_{i,j,b}, Q^{Fdr}_{i,j,b}$	active and reactive power flow from i to j , p.u.
$p^{PEVs}_{i,b}$	power drawn by PEV in smart mode, p.u.
$P_{i,b}, Q_{i,b}$	real and reactive power imported by LDC via substation, p.u.
$p^{UN}_{i,b}$	unserved power at bus i and load block b , p.u.
Q^{CCap}	capacity of shunt capacitor, p.u.
$Q^C_{i,b}$	reactive power injected by capacitor i , p.u.
$S^{NFdr}_{i,j}$	capacity added to feeder i – j , p.u.
$S^{NSS}_{ss}, P^{NSS}_{ss}$	capacity added to substation, p.u.

$S_{ss,b}$	apparent power imported via substation, p.u.
$V_{i,b}$	voltage magnitude at bus i and load block b , p.u.
z^{DG}_i	binary decision on DG investment, (0/1)
$z^{Fdr}_{i,j}$	binary decision on feeder upgrade, (0/1)
z^{SS}_{ss}	binary decision on substation upgrade, (0/1)
$\theta^{Fdr}_{i,j,b}$	power angle of the flow from i to j , rad
$\delta_{i,b}$	voltage phase angle at bus i and load block b , rad

penetration. In addition to traditional plan decisions, these issues need be examined, and they form the core of future distribution system planning.

The transition towards the smart grid may have a significant impact on distribution networks. A smart distribution grid involves the use of information and communication technology, system automation technologies, DR, and PEV loads. Customers play an important role in smart grids by introducing flexibility in electricity usage, and new DR programs are being developed to achieve the desired load shaping [2].

There is a large body of literature that focus on traditional distribution system planning which involves optimal feeder design, configuration, substation design, etc. [1], while many recent works consider the problem of optimal DG sizing and siting in distribution systems [3]. In [4], a multi-objective optimization problem is developed to solve the expansion planning problem including DG units. While in [5,6], multi-year optimization frameworks for distribution planning including DG units are proposed to determine the optimal period, capacities, and location of upgrades. In [7–9], two-stage heuristic approaches are proposed to determine the optimal timing, locations, and capacities of feeders, and DG units. In [10] a multi-year distribution system planning framework is proposed that uses a fundamental economic criterion, the ‘cost-benefit analysis’, in a heuristic selection process of plan options, starting from the terminal year and propagating backwards to the initial year, to arrive at a plan solution. A multi-objective optimization model is proposed in [11] to determine the optimal DG sizes and locations while minimizing the LDC’s cost and maximizing the DG owners’ benefits. In [12], a heuristic approach is proposed to allocate DGs based on loss reduction and voltage improvement sensitivity of the system.

When system upgrades are essentially driven by the continuously increasing demand, capacitors can play a vital role in deferring the need for upgrades. They can improve voltage profiles, reduce feeder losses and network congestion by reducing the reactive power flows on the feeders. Coordination of capacitor and DG placement can therefore maximize the savings from loss reduction in distribution networks. Although, in the literature, there are several studies that consider the coordination of capacitors and DG placements [13–15], very few researchers have considered the comprehensive range of options such as capacitors, DGs, feeders and substations [16], while none have considered PEV charging loads and DR options. In [13–15], a multi-objective optimization problem are formulated for simultaneous allocation of DGs and capacitors in distribution networks.

It is evident that the impact of PEV charging on the distribution grid will be significant as PEV penetration increases in the coming years. Several researchers have considered the integration of PEVs into distribution system operation [17–21], smart charging and discharging strategies [22–26], and the interaction between PEV charging load and renewable resources [27–30]. In some recent works, researchers have examined the allocation of parking lots equipped with PEV charging facility to minimize

reliability indices [31] and considering the presence of renewables in [32]. It is noted that most of the works have focussed on the impact of PEV charging loads on distribution system operations while only a few on planning. In [33], a multi-objective optimization model is proposed which considers the optimal PEV penetration, DG units and feeder upgrades in a distribution system.

DG ownership by LDCs is a critical issue that need be examined, and which can provide many benefits to LDCs that can regulate and decide on the DG siting, sizing and operational decisions [34]. In the present work, it is therefore assumed that the LDC is allowed to invest in DG units at strategic locations on the grid in order to defer network upgrade costs and reduce its cost of purchasing peak-hour energy from the external grid. This is in line with the prevailing regulations in Ontario, Canada [35]. As per the amendments of Ontario Energy Board [35], the LDCs in Ontario are permitted to own and operate generation and storage facilities, and are required to treat these in the same manner as they would treat facilities owned by third parties. Therefore, DG units have been included as plan decision variables alongside other distribution system components, as part of the LDC's planning problem.

So far, planning of distribution systems considering all these resources, namely, DGs, capacitors, feeders and transformers, and including PEV charging loads (both uncontrolled and smart) and DR mechanisms, has not been reported in the literature. Distribution system planning should therefore be carried out in such a way that the resultant plan provides a reliable and cost effective service to customers while satisfying constraints. Considering all these resources is an important requirement of the planning process that can impact the outcomes significantly. As a result, it is important to develop and implement effective planning strategies that properly address these issues.

The aim of this work is to present a comprehensive distribution planning framework that is suited to take into account the changing paradigms of today, in the context of smart grids, while also guaranteeing an acceptable level of adequacy. This paper builds upon the work reported in [10], with several model improvements and features that allow for important policy conclusions. Thus, the main contributions of this paper, over and above the previous works on this subject including [10], are as follows:

- A comprehensive multi-year distribution system planning framework is proposed to simultaneously determine the optimal size, site, and period of commissioning of substations, DGs, feeders, and capacitors.
- A novel adequacy check module is introduced in the planning process which ensures that a desired level of reserve margin is maintained in the distribution system.
- In addition to considering various new and additional features in the planning model such as capacitors, and PEV charging loads, the proposed planning framework also considers DR as a plan option to mitigate the impact of PEV charging loads, and particularly when the plan is constrained by budgetary limits.
- The uncontrolled PEV charging load profile is estimated considering real transportation data including driver behaviour, arrival time, mileage, and other information for inclusion in the distribution planning problem.
- A novel set of PEV operational constraints are included in the distribution planning model, to capture the impact of PEV smart charging loads.

The structure of the paper is as follows. In Section 2, the mathematical model of the proposed multi-year planning problem is presented and this is followed by a description of the proposed solution approach in Section 3. In Section 4, the two test systems

considered for analysis are described and the results are presented in Section 5. Conclusions are drawn in Section 6.

2. Smart distribution system planning model

2.1. Smart-DSPLAN

The first proposed generic mathematical model for long-term distribution planning, considering the comprehensive set of decisions and issues discussed earlier, is referred to as, Smart-DSPLAN, and is presented below.

2.1.1. Objective function

The objective function (J) aims to minimize the annualized cost of capital and operation of the LDC.

$$\begin{aligned}
 J = & \sum_{i \in N} \left(C^{DG,F} P_i^{DG, Cap} + \sum_{b \in B} (C^{DG,O} P_{i,b}^{DG} Hr_b) \right) \\
 & + \sum_{i \in SS} \left(C^{SS,F} Z_i^{SS} + C^{SS,V} S_i^{SS} + \sum_{b \in B} (\rho_b P_{i,b} Hr_b) \right) \\
 & + \sum_{i,j \in N: \exists(i,j)} (C^{Fdr,F} G_{e_{i,j}} L_{e_{i,j}} Z_{i,j}^{Fdr} + C^{Fdr,V} S_{i,j}^{NFdr}) + \sum_{i \in N} (C^C Q_i^{C_{Cap}}) \\
 & + \sum_{i \in N, b \in B} (C^{DR} P_{i,b}^{DR} Hr_b) \quad (1)
 \end{aligned}$$

The first line of (1) includes the capital and operating cost of the candidate DG units. The second line includes the engineering, procurement, and construction (EPC) cost and the variable component of the capital cost to upgrade the substation, and payment towards purchased power by the LDC. The third line represents the EPC cost, and the variable component of the capital cost to upgrade the feeders, the last line denotes the capital cost of candidate capacitors, and the payment made by the LDC to DR customers. The associated operational and planning constraints are discussed next.

2.1.2. Power flow equations

The injected power at a bus is the power from the substation and DG units, net of the load, uncontrolled and smart PEV charging loads, and DR; and is governed by the traditional ac power flow equations:

$$\begin{aligned}
 P_{i,b} + P_{i,b}^{DG} - P_{i,b}^{PEVs} - P_{i,b}^{DR} + P_{i,b}^{DR} = \\
 \sum_{j \in N} V_{i,b} V_{j,b} Y_{i,j} \cos(\theta_{i,j} + \delta_{j,b} - \delta_{i,b}) \quad \forall i, b \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 Q_{i,b} + Q_{i,b}^C - Q_{i,b} = \\
 - \sum_{j \in N} V_{i,b} V_{j,b} Y_{i,j} \sin(\theta_{i,j} + \delta_{j,b} - \delta_{i,b}) \quad \forall i, b \quad (3)
 \end{aligned}$$

2.1.3. Feeder capacity limits

Power flow through any distribution feeder must comply with the thermal capacity of the feeder. This limit also takes into consideration the new investments in feeder upgrades.

$$\begin{aligned}
 -V_{i,b}^2 Y_{i,j} \cos \theta_{i,j} + V_{i,b} V_{j,b} Y_{i,j} \cos(\theta_{i,j} + \delta_{j,b} - \delta_{i,b}) \leq \\
 (S_{i,j}^{Fdr, Cap} + S_{i,j}^{NFdr}) \cos \theta_{i,j}^{Fdr} \quad \forall (i, j) \in N: \exists(i, j), \forall b \quad (4)
 \end{aligned}$$

$$V_{i,b}^2 Y_{i,j} \sin \theta_{i,j} - V_{i,b} V_{j,b} Y_{i,j} \sin(\theta_{i,j} + \delta_{j,b} - \delta_{i,b}) \leq \quad (5)$$

$$(S_{i,j}^{FdrCap} + S_{i,j}^{NFdr}) \sin \theta_{i,j}^{Fdr} \quad \forall (i, j) \in N : \exists (i, j), \forall b \quad (6)$$

$$S_{i,j,h}^{NFdr} \leq M z_{i,j}^{Fdr} \quad \forall (i, j) \in N : \exists (i, j), \forall b \quad (6)$$

In (6), M is a sufficiently large number often called the “big M ”, which renders the constraint (6) nonbinding. When $z_{i,j}^{Fdr} = 0$, $S_{i,j}^{NFdr}$ is zero while when $z_{i,j}^{Fdr} \neq 0$, the large value of M allows sufficient room for selection of new feeder capacity, and the value of M ensures that (6) is satisfied. The binary variables $z_{i,j}^{Fdr}$ act as a “switch” on the continuous constraints, appearing with a large number greater than or equal to the maximum allowable feeder upgrade capacity.

2.1.4. Substation capacity limits

These constraints ensure that the total power delivered by the substation transformer is within the substation capacity limit. These limits take into consideration new investments in substation upgrades.

$$S_{ss,b} \leq S_{ss}^{SSCap} + S_{ss}^{NSS} \quad \forall b \quad (7)$$

$$S_{ss}^{NSS} \leq M z_{ss}^{SS} \quad (8)$$

2.1.5. DG capacity limits

The power generated by a DG unit is limited by the DG capacity (9). The installed capacity of DG is limited by the maximum allowable DG size (10). Constraint (11) limits the number of allowable DG units. The maximum DG penetration is limited by the minimum load (including PEV charging load) plus 60% of maximum substation rating in order to limit the maximum reverse power flow over the transformer, as given in (12)[36].

$$P_{i,b}^{DG} \leq P_i^{DGCap} \quad \forall i, b \quad (9)$$

$$P_i^{DGCap} \leq DG^{CapMax} z_i^{DG} \quad \forall i \quad (10)$$

$$\sum_{i \in N} z_i^{DG} \leq DG^{NMax} \quad (11)$$

$$\sum_{i \in N} P_{i,b}^{DG} \leq \sum_{i \in N} (Pd_{i,b} + P_{i,b}^{PEV}) + 0.6 P_{i,b} \quad \forall b \quad (12)$$

2.1.6. Capacitor limits

These constraints ensure that the installed capacitor size is limited by a specified maximum allowable capacitor size (13). The reactive power injected by a capacitor must be less than the installed capacity (14).

$$Q_i^{Cap} \leq Q_i^{CMax} \quad \forall i \quad (13)$$

$$Q_{i,b}^C \leq Q_i^{Cap} \quad \forall i, b \quad (14)$$

2.1.7. PEV smart charging constraints

These constraints determine the optimal PEV charging schedule by ensuring that the total energy required by PEVs is equal to their daily energy needed to charge the battery (E); and the power drawn by PEVs is within the charging level. It is to be noted that E is calculated based on the average mileage driven by a vehicle as determined from the National Household Travel Survey data (NHTS) [37].

$$\sum_{b \in B} P_{i,b}^{PEV} = NH_i \cdot E \cdot PEV^{Pen} \quad \forall i \quad (15)$$

$$P_{i,b}^{PEV} \leq NH_i \cdot P_{i,b}^{PEVCap} \cdot PEV^{Pen} \quad \forall i, h \quad (16)$$

2.1.8. Capacity adequacy limit

This constraint ensures the installation of enough capacity in the system so that supply can be maintained during peak hours in case of resource failure.

$$P_{ss}^{SSCap} + P_{ss}^{NSS} + \sum_{i \in N} P_i^{DGCap} \geq (1 + \gamma) \sum_{i \in N} (Pd_{i,peak}) \quad (17)$$

2.1.9. DR constraints

These constraints impose a limit on the DR at each time block and bus in the system.

$$P_{i,b}^{DR} \leq \alpha_0 Pd_{i,b} \quad \forall i, b \quad (18)$$

2.1.10. Voltage limits

These constraints ensure that the voltage magnitude at a bus is within the allowable limits.

$$V^{Min} \leq V_{i,b} \leq V^{Max} \quad \forall i, b \quad (19)$$

2.1.11. Budget limit-general

This constraint imposes a limit on the capital spending and consequently how much capacity the LDC can invest. The first term of (20) is the capital cost of DG units, the second term is the EPC cost and the variable component of the capital cost to upgrade the substation. The third term is the EPC cost and the variable component of the capital cost to upgrade the feeders. The capital cost of capacitors is presented in the last term. All these costs together, must be within the budget limit, BL .

$$\begin{aligned} & \sum_{i \in N} (C^{DG.F} P_i^{DGCap}) + \sum_{i \in SS} (C^{SS.F} z_i^{SS} + C^{SS.V} S_i^{NSS}) \\ & + \sum_{i,j \in N: \exists(i,j)} (C^{Fdr.F} Ge_{i,j} Le_{i,j} z_{i,j}^{Fdr} + C^{Fdr.V} S_{i,j}^{NFdr}) \\ & + \sum_{i \in N} (C^C Q_i^{Cap}) \leq BL \end{aligned} \quad (20)$$

2.1.12. Budget limit on capacitor banks

This constraint imposes a limit on spending that the LDC can make on new capacitor banks.

$$\sum_{i \in N} C^C Q_i^{Cap} \leq CBL \quad (21)$$

It is to be noted that the capital cost of capacitor banks are relatively smaller compared to that of DG units, and if a separate budget limit for capacitors is not imposed, the planning model will tend to over-select the number of capacitors, at the cost of DG units, which is undesirable. To alleviate this problem, a separate budget is therefore allocated to capacitor banks, without any loss of generality.

2.2. Smart-DSPLAN2

The second proposed mathematical model, referred to as, Smart-DSPLAN2, discussed earlier, is presented below.

The objective function (J_2) aims to minimize the annualized cost of capital and operation of the LDC.

$$\begin{aligned}
 J_2 = & \sum_{i \in N} \left(C^{DG.F} P_i^{DG.Cap} + \sum_{b \in B} (C^{DG.O} P_{i,b}^{DG} H_{r_b}) \right) \\
 & + \sum_{i \in SS} \left(C^{SS.F} + C^{SS.V} S_i^{SS} + \sum_{b \in B} (\rho_b P_{i,b} H_{r_b}) \right) \\
 & + \sum_{i,j \in N: \exists(i,j)} (C^{Fdr.F} G e_{i,j} L e_{i,j} + C^{Fdr.V} S_{i,j}^{NFdr}) + \sum_{i \in N} (C^C Q_i C_{cap}) \\
 & + \sum_{i \in N, b \in B} (C^{DR} P_{i,b}^{DR} H_{r_b}) + \sum_{i \in N, b \in B} (C^{UN} P_{i,b}^{UN} H_{r_b}) \quad (22)
 \end{aligned}$$

The associated operational and planning constraints are same as Smart-DSPLAN constraints with some differences as follows:

- Constraints (6), (8), (10), (11), (13), (17), (20) and (21) are excluded.
- Constraints (2), is modified to include unserved power variable as follows:

$$\begin{aligned}
 P_{i,b} + P_{i,b}^{DG} - P_{i,b}^{PEVS} - P_{i,b}^{PEVUNC} - \\
 P_{d,i,b} + P_{i,b}^{DR} + P_{i,b}^{UN} = f(V_{i,b}, \delta_{i,b}) \quad \forall i, b \quad (23)
 \end{aligned}$$

- The upgrades capacities are now considered as a fixed decisions.

3. Proposed approach to solution of planning problem

The proposed approach to determine the optimal upgrade plan combines a bi-level procedure as follows:

Level-1 (ADEQ-OPTSELECT): determines an appropriate reserve margin (γ) and selects the optimal size and location of distribution system upgrades that are required to be in place, at the plan terminal year.

Level-2 (ADEQ-OPTPERIOD): determines the optimal period of commissioning, for the upgrades selected in Level-1.

A back-propagation of model solution, starting from the plan terminal year T and ending at the first year, is proposed. Within this framework, a novel adequacy check is introduced that involves post-processing the plan decisions at both the levels to ensure that the target plan adequacy level is satisfied for each year (Fig. 1). The details of the proposed framework are discussed next.

3.1. Level-1: ADEQ-OPTSELECT

In this stage the optimal plan decisions for the plan terminal year are determined. The step-by-step procedure is discussed as follows.

1. Start with an initial value of reserve margin γ and consider all LDC components as candidates for upgrade set $\{L\}$.
2. Execute Smart-DSPLAN model, considering set $\{L\}$, to determine the optimal planning decisions, set $\{H\}$, which provides continuous values of capacities.
3. Standardize the upgrade capacities of $\{H\}$ and consider them as fixed decisions in the Smart-DSPLAN. Also, now include an unserved power variable in the demand-supply balance (23) with an associated high cost of unserved power in the objective function (22). The modified version is now referred to as Smart-DSPLAN2.
4. Execute Smart-DSPLAN2 to calculate the Benefit-Cost-Ratio (BCR) of each upgrade in set $\{H\}$ by uninstalling an upgrade, one at a time, from $\{H\}$, and calculate the marginal benefit of each upgrade. The marginal benefit for an upgrade is obtained from the difference in the objective function (22) before and after removing the element. The BCR for an upgrade is calculated by dividing the marginal benefit by the total cost of the upgrade as follow:

$$BCR_{i,Type} = \frac{MB_{i,Type}}{TC_{i,Type}} \quad (24)$$

where $MB_{i,Type}$ is the marginal benefit obtained from the difference in the objective function (J) before and after removing a specific upgrade; $TC_{i,Type}$ is the total cost of installing an upgrade; and $Type$ is the type of upgrade (substation, DG, capacitor or feeder)

5. Select all upgrades with $BCR > 1$, and form a set $\{H1\}$, also construct a set $\{R\}$ with rejected upgrades having $BCR < 1$ and solve Smart-DSPLAN2 with set $H1$ to check if the above selected upgrades satisfy the system constraints and there is no unserved power.
6. If YES, go to next step. If NO, modify set $\{L\}$ with all buses except those in set $\{R\}$. If the selected upgrades from $\{H1\}$ do not satisfy system constraints, and all upgrades from set $\{L\}$ have been tested for $BCR > 1$, then choose upgrades with the highest BCR from the rejected set $\{R\}$. If otherwise, go to Step 2.
7. Develop a capacity outage probability distribution table. Calculate the Loss of Load Expectation (LOLE) [38] for the obtained plan $\{H1\}$ and check if LOLE is less than the specified value.
8. If YES, the final solution is obtained. If NO, revise γ as: $\gamma = \gamma + \Delta\gamma$, which means the reserve margin, and hence supply capacity (DG

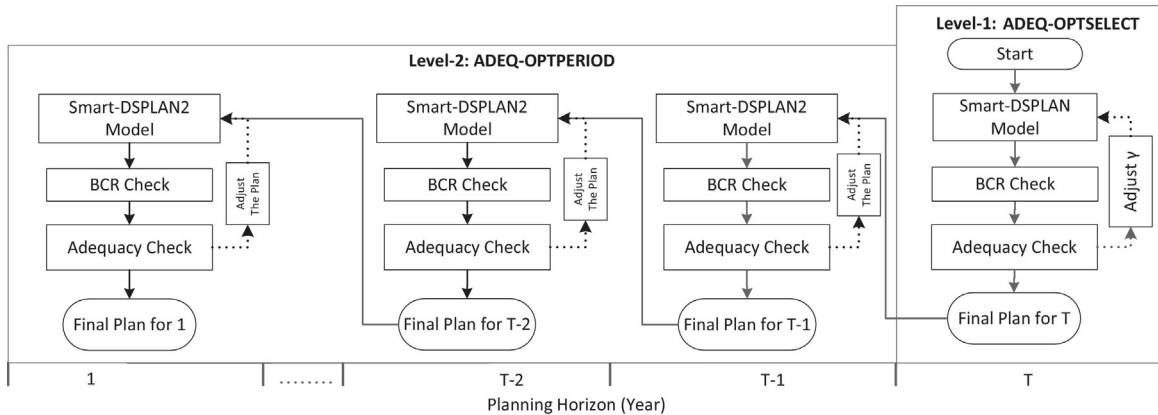


Fig. 1. Architecture of the proposed approach to solution of planning problem.

and substation), is appropriately increased to meet the capacity adequacy limit in (17), and go to Step 2.

It is to be noted that the choice of the starting value of γ has no significant impact on the distribution plan. If the initial choice of this margin is too far off, the number of iterations required for convergence will be more. But since this is a long-term planning problem, solved 10 years in advance, computational time is not a critical issue, and a few additional iterations will not affect the plan. Moreover, there is no unique starting value of γ that is universally appropriate for all distribution planning problems. This is because the LOLE depends on a number of factors such as the magnitude and the duration of each load block and the number of load blocks considered, the FOR, and the capacity of the generation units. Any knowledgeable guess by the distribution system planner is sufficient; in this work, a value of $\gamma = 0.15$, which is the reserve margin in year-0 is used; and γ is increased in steps of 0.01 p.u. per iteration.

The final plan for year T is now passed on to Level-2 wherein through the back propagation process, the optimal year of commissioning of the plan upgrades are obtained.

3.2. Level-2: ADEQ-OPTPERIOD

The ADEQ-OPTPERIOD starts from year $(T - 1)$, and propagates back to the first year of the plan horizon to find the optimal set of upgrades for each year of the plan.

Two essential verification tests are used at this level to determine the optimal set of upgrades for each year. The first test, BCR, is used to reject the non-beneficial upgrades from that year and earlier. While the second, is the adequacy check which ensures that the selected upgrades meet a specified adequacy level.

In each year of the plan, BCR and LOLE are calculated starting from year $(T - 1)$. Based on the BCR check, any upgrade with a BCR of less than unity will be a possible candidate for rejection. If the selected upgrade does not satisfy the system constraints and the adequacy level, the rejected upgrades will be re-selected one by one based on their BCR, until the system constraints are satisfied, and the adequacy level is met. The step-by-step procedure of Level-2 is discussed as follows:

1. Set initial value $t = T - 1$.
2. Solve Smart-DSPLAN2 for year t and calculate BCR for all upgrades.
3. Reject upgrades with $BCR < 1$ from $\{H1\}$ and form rejected set $\{R2\}$. It is to be noted that the upgrades that are rejected, are only for year t and earlier, implying that these upgrades are made in year $t + 1$.
4. Check if selected upgrades in $\{H1\}$ satisfy the system constraints. If YES, go to step-5. If NO, re-select the upgrade with the highest BCR from $\{R2\}$ and perform step-4 again.
5. Develop a capacity outage probability distribution table and calculate the Loss of Load Expectation (LOLE) for the obtained plan $H1$ and check if LOLE is less than the specified value. If NO, re-select the upgrade with the highest BCR from $\{R2\}$ and perform step-5 again.
6. Modify set $\{H1\}$ with all upgrades that are selected for year t .
7. Update $t = t - 1$. If $t \neq 0$, go to Step-2. ELSE, the final plan is obtained.

4. Test system

The first test system under study comprises 33 buses in radial configuration [39]. The main substation at bus-1 has two transformers of 15 MVA each, and one of 16 MVA. The total system peak demand is 37 MW in year-0. The second test system is the IEEE

Table 1
Capital cost of utility resources [5,10].

Element	Fixed cost	Variable cost (\$/MVA)
Feeder	150,000\$/km	1000
Substation	\$200,000	50,000
Gas turbine DG	–	825,000
Capacitor	–	50,000

Table 2
Electricity market price [45].

Load block (b)	1	2	3	4	5	6	7
Number of hours	5	4	4	4	3	3	1
LSF	0.4	0.5	0.6	0.7	0.8	0.9	1
ρ (\$/MWh)	10	15	22	32	48	70	103

69-bus radial distribution feeder comprising one main branch and seven laterals [40]. The main substation at bus-1 has two transformers of 1.5 MVA each, and one of 2.2 MVA. The total system peak demand is about 3.8 MW in year-0.

For both systems, the demand is assumed to grow at 3% annually, all the substation transformers are assumed to have a force outage rate (FOR) of 0.02 [41], and the feeder segments are assumed to be 1 km long, $Ge = 0.4$ [5]. Table 1 provides the capital costs of the resources available for planning. Details of various other parameters used for the studies are taken from [5,10]. Assuming a fuel consumption of 300 m³/h and a gas price of 0.14 \$/L, the operating cost of the DG units are obtained to be 42 \$/MWh [42], and have a FOR of 0.05 [41]. A budgetary limit of \$10 million is imposed. The targeted LOLE is assumed to be 2.8 h/year and the unserved energy cost is 1000 \$/MWh [43]. The load profile is represented by load duration curves which are constructed by re-arranging the chronological load curve in descending order of magnitude. It determines the number of hours per day when the load is greater than any particular amount [44]. Electricity price or the price at which the LDC imports power through the substation are specified in terms of seven load blocks, as shown in Table 2, using load scaling factors (LSF). The average Hourly Ontario Energy Price (HOEP) in 2012 is considered in this study to be the expected electricity price for the plan period [45]. Note that the planning problem addressed in this paper is from the perspective of an LDC, and hence the issues of electricity market prices, bidding, and risk management, are not considered. The price at which the LDC purchases power from the external grid is assumed to be known *a priori*, and the LDC does not participate in the wholesale electricity market to purchase this power, but has a standing power purchase contract.

In the present work, it is assumed that the DG units are dispatchable, and are natural-gas based; therefore siting or sizing of DG units is not constrained by the availability of resources and can be optimally determined from the model as per system requirements. In this work, the DG units are assumed to operate at unity power factor, as per [7,33], however, the proposed algorithm is generic and computationally fast to handle DG reactive power support capability. The Smart-DSPLAN is a mixed-integer non-linear programming model, solved using the COINBONMIN solver; while Smart-DSPLAN2 is a non-linear programming model, solved using the SNOPT solver, and are coded in GAMS.

5. Case studies

5.1. Case-1: Base Case; 32 Bus System

In this case, the PEV charging load and DR are not considered, accordingly (17), (18) and (20), and the associated variables are excluded from the Smart-DSPLAN model. The results are divided into two parts; the first part determines the optimal upgrade plan

Table 3
Reserve margin, added capacity and LOLE.

γ	Added capacity (MVA)	LOLE (h/year)	Target LOLE (h/year)	Adequacy level met?
0.15	21.4	3.47	2.8	No
0.2	24.1	3.17	2.8	No
0.21	24.7	2.29	2.8	Yes

for the terminal year. Using this output, the year of commissioning of selected upgrades is obtained in the second part.

5.1.1. ADEQ-OPTSELECT

The proposed schematic discussed in Section 3.1 is used to determine the optimal γ simultaneously with the optimal size and location of LDC component upgrades in the plan terminal year (T). Table 3 presents the outcome of the iterative process to arrive at the optimal value of γ .

After three iterations, it is found that with $\gamma=0.21$, LOLE is obtained as 2.29 h/year which meets the targeted adequacy level. The step-by-step outcome of the ADEQ-OPTSELECT process for $\gamma=0.21$ is thereafter presented in Table 4. In the first iteration, set {H} of selected upgrades comprises the upgrade of 1 substation, 5 DG units, 6 capacitors and 5 feeders. The BCRs for each selected upgrade is calculated and found to be greater than unity except for DG at bus 32, which is hence rejected. In the second iteration, a new set {H} is obtained. After calculating BCRs for each selected upgrade, all of them are found to have a $BCR > 1$, and hence, the optimal plan for the terminal year is obtained, denoted by set {H1}.

Table 4
Step-by step outcome of ADEQ-OPTSELECT with $\gamma=0.21$.

Itr.	Set H of upgrades	Size, MW	BCR	Set H1	Rejected set R
1	Substation	15.4	5.56	Substation	
	DG #13	2.2	1.1	DG #13	
	DG #24	0.7	1.4	DG #24	
	DG #29	3.5	1.2	DG #29	
	DG #31	2.3	1.1	DG #31	
	DG #32	0.7	0.8	–	
	Capacitor #18	0.1	10.22	Capacitor #18	
	Capacitor #19	0.3	15.09	Capacitor #19	
	Capacitor #20	0.3	15.12	Capacitor #20	DG #32
	Capacitor #21	0.3	15	Capacitor #21	
	Capacitor #31	0.3	110.81	Capacitor #31	
	Capacitor #32	0.2	106.02	Capacitor #32	
	Feeder 1–2	6.2	303.73	Feeder 1–2	
	Feeder 2–22	1.5	76.04	Feeder 2–22	
	Feeder 3–4	1.1	44.15	Feeder 3–4	
	Feeder 26–27	0.7	43.33	Feeder 26–27	
	Feeder 28–29	0.6	55.66	Feeder 28–29	
2	Substation	15.9	6.3	Substation	
	DG #17	0.7	1.3	DG #17	
	DG #21	0.4	1.2	DG #21	
	DG #24	0.7	1.4	DG #24	
	DG #29	3.8	1.3	DG #29	
	DG #31	3.2	1.2	DG #31	
	Capacitor #13	0.1	2.68	Capacitor #13	
	Capacitor #14	0.1	2.67	Capacitor #14	
	Capacitor #15	0.1	2.71	Capacitor #15	
	Capacitor #16	0.1	2.74	Capacitor #16	None
	Capacitor #17	0.3	26.65	Capacitor #17	
	Capacitor #30	0.3	126.86	Capacitor #30	
	Capacitor #31	0.3	126.99	Capacitor #31	
	Capacitor #32	0.2	125.23	Capacitor #32	
	Feeder 1–2	6.4	350.08	Feeder 1–2	
	Feeder 2–22	1.5	76.04	Feeder 2–22	
	Feeder 3–4	1.1	48.55	Feeder 3–4	
	Feeder 26–27	0.2	8.6	Feeder 26–27	
	Feeder 28–29	0.2	19.08	Feeder 28–29	

5.1.2. ADEQ-OPTPERIOD

At this level, the final selected upgrades from ADEQ-OPTSELECT, i.e., set {H1}, are considered again. The LOLE for the upgraded system and BCR corresponding to each component of {H1} are determined starting from year ($T-1$), and the optimal upgrade plan for each year is obtained. In the present case study, first, the BCR is calculated for year 9, considering the selected upgrades {H1} from Table 4. It is found that three feeders have $BCR < 1$ and hence these are possible candidates for rejection and the updated plan for year 9 is checked using Smart-DSPLAN2. It is found that the system is feasible and there is no unserved power resulting from these rejected upgrades. The LOLE considering the updated plan for year 9 is 1.57 h/year and meets the targeted adequacy level. Therefore, the three feeders are removed from year 9 and earlier years, and installed at year 10. Table 5 presents the optimal period of commissioning of each upgrade as obtained from ADEQ-OPTPERIOD.

In Table 6, the step-by-step procedure of ADEQ-OPTPERIOD, considering a sample year, year 6 is presented. The revised set {H1} at the beginning of year 6 is as follows: {Substation, DG #29, #31, all Capacitors, Feeder 1–2, 2–22}. After calculating BCR for all these upgrades, it is found that DG #29 and #31 have $BCR < 1$. When Smart-DSPLAN2 is executed without these upgrades, the system encounters unserved energy. As explained in the step-by-step procedure of ADEQ-OPTPERIOD, DG #29 having the highest BCR amongst the rejected upgrades, is re-introduced at the end of iteration-1. Following the same procedure, upgrade of DG #31 is rejected in iteration-2, but system LOLE is found to be higher than the targeted adequacy level and hence DG #31 is re-introduced. In iteration-3, with no rejected upgrades LOLE is calculated and it is found that the plan meets the targeted adequacy level for year-6.

Table 5
Case-1 results of ADEQ-OPTPERIOD.

Year	Upgrades size (MW) and site (Bus)	LOLE (h/year)
10	(1.1) Fdr 3–4, (0.2) Fdr 26–27, and (0.2) Fdr 28–29	2.29
9	(0.7) DG #17, and (0.7) DG #24	1.57
8	(0.4) DG #21	1.55
7	–	1.52
6	(3.2) DG #31	1.46
5	(1.5) Fdr 2–22	1.9
4	(6.4) Fdr 1–2	1.39
3	–	1.39
2	–	1.18
1	(15.9) Substation, (3.8) DG #29, Capacitors: (0.1) #13, (0.1) #14, (0.1) #15, (0.1) #16, (0.3) #17, (0.3) #30, (0.3) #31, and (0.2) #32	0.48

5.2. Case-2: PEV penetration; 32-bus test system

The impact of PEV charging load on distribution system planning is examined in this section. Two case studies are performed, smart charging and uncontrolled charging, to evaluate their impact on the plan. It is assumed that the entire load is residential and by the plan terminal year there would be one PEV every two houses; i.e. 50% PEV penetration. The number of PEVs connected at a bus is calculated based on a typical average hourly load of a house, 2.08 kW [24]. The SAE J1772 standards defines two residential charging levels (Level-1 and Level-2) with a third ac level and a dc level for PEVs [46], as presented in Table 7. In our work we have considered Level-2 (low) charging with a charging efficiency of 85%. PHEV60 is considered in this case study, with a battery capacity of 15.9 kWh [47].

For this case there is a need for a real and detailed travel dataset, for example the NHTS [37], which is used herein to estimate the PEV driving pattern parameters such as daily mileage driven and the probability of latest home arrival time. The PEV charging load profile is developed considering the start of charging time, the energy required at each load bus, and the charging duration.

5.2.1. Case-2a: smart charging

In this case, the LDC is assumed to have a control on PEV charging schedules. Accordingly, the DR related variable and constraint (20) are excluded from the Smart-DSPLAN model. The approach presented in Section-III is executed and results from ADEQ-OPTSELECT are obtained. It is seen that the targeted adequacy level is now met with a higher value of $\gamma = 0.31$, as against $\gamma = 0.21$ without PEVs; this is attributed to the increased system demand from PEV charging. The optimal plan outcome of this level comprises one substation upgrade, five DG units, five capacitors and three feeder upgrades; and one iteration is needed to arrive at the final plan.

At level-2 (ADEQ-OPTPERIOD), PEV penetration over the plan period is assumed to increase exponentially; starting at approximately 1.5% in year 1 and touching 50% in the plan terminal year. The plan outcome of this level (Table 8) shows that the optimal charging of PEVs occurs during the base load (load block = 1) because of the prevailing low market price shown in Table 2.

Table 6
ADEQ-OPTPERIOD step-by-step procedure for year 6.

Iteration of ADEQ-OPTPERIOD	Rejected upgrades	Unserved power	Re-introduce upgrade?	LOLE
1	DG #29 DG #31	Yes	DG #29	–
2	DG #31	No	DG #31	3.96
3		No		1.09

Table 7
PEV charging level characteristics.

Type	Power level
Level 1: 120 VAC	1.2–2.0 kW
Level 2 (low): 208–240 VAC	2.8–3.8 kW
Level 2: (high): 208–240 VAC	6–15 kW
Level 3: 208–240 VAC	>15–96 kW
Level 3: DC charging: 600VDC	>15–240 kW

Table 8
Case-2a results of ADEQ-OPTPERIOD.

Year	Upgrades size (MW) and site (Bus)	PEV Loading in MW
10	(0.4) Fdr 22–23	22.21
9	(0.4) DG #24 and (1.9) #29	15.02
8	(0.4) DG #21	10.16
7	(2.5) DG #14	6.87
6	–	4.64
5	(1.8) Fdr 2–22	3.14
4	(5.5) Fdr 1–2	2.12
3	–	1.44
2	–	0.97
1	(20.9) Substation, (4.6) DG #30, Capacitors: (0.3) #28, (0.3) #29, (0.3) #30, (0.3) #31, and (0.3) #32	0.66

Note: PEV loading occurs during the base load (load block = 1) at each year of the plan.

5.2.2. Case-2b: uncontrolled charging

In this case, the impact of uncontrolled PEV charging on the required distribution system upgrades is evaluated. The PEV smart charging related constraints (17) and (18) and associated variables are now excluded from the Smart-DSPLAN model, while the DR variables and constraint (18) is now included. A high DR cost of 1000 \$/MWh is assumed, and a budgetary limit of \$15 million is imposed for this case study. The DR option is included in order to negotiate a distribution plan when budgetary constraints are imposed on the planner. In the second level (ADEQ-OPTPERIOD), the maximum forecasted DR penetration is assumed to be 1% in year 1, which increases exponentially; rising to 20% in the plan terminal year.

The PEV charging load is simulated in two ways to mimic the behaviour of PEV customers. The first group is assumed to start charging immediately after arriving home while the second group's behavior is modelled considering a Poisson distribution to mimic those drivers' behavior who arrive at on-peak electricity price hours and wait until the onset of off-peak price to plug-in their vehicles. The charging delay (λ) is assumed considering Ontario's Time-of-Use (TOU) electricity price; for vehicles arriving during off-peak hours (7 PM to 7 AM), $\lambda = 0.1$ h, while for on-peak arrivals, λ is the wait time between their arrival and onset of off-peak TOU price. The energy required by a PEV is calculated based on its mileage driven. The charging duration is calculated from the power drawn at a given charging level, the charging efficiency, and the energy required. The resulting uncontrolled PEV charging profile is then combined with the base demand to form the total demand profile as shown in Fig. 2.

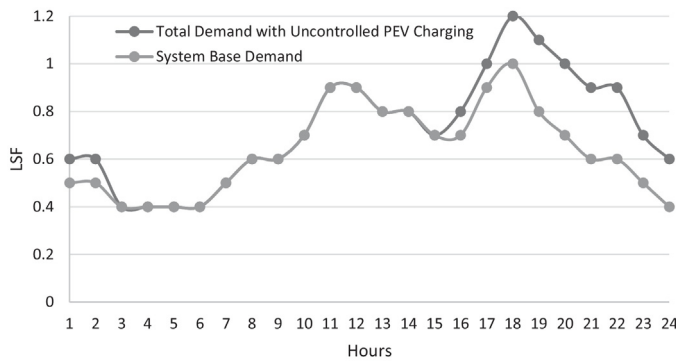


Fig. 2. System demand without and with uncontrolled PEV charging.

Table 9

Case-2b: results of ADEQ-OPTPERIOD.

Year	Upgrades size (MW) and site (Bus)	DR capacity (MW)
10	(1.2) Fdr 2–3, (2.8) Fdr 3–4, (1.6) Fdr 4–5, (1.7) Fdr 26–27, and (1.1) Fdr 27–28	1.21
9	(0.1) Capacitor #21, and (2.9) DG #29	0.266
8	(0.9) DG #32	0.191
7	(11.5) Fdr 1–2, (0.6) Fdr 1–18, (3.2) Fdr 22–23, (1.6) Fdr 23–24, (2) Fdr 28–29 and (1.7) DG #28	0.137
6	–	0
5	(5) Fdr 2–22	0
4	(5) DG #12	0
3	–	0
2	–	0
1	Capacitors: (0.2) #28, (0.3) #29, (0.3) #30, (0.3) #31, and (0.3) #32, (5) DG #30, and (23.3) Substation	0

The proposed approach presented in Section 3 is executed and results from ADEQ-OPTSELECT are obtained, the targeted adequacy level is met with $\gamma = 0.54$. It is noted that the optimal γ is higher compared to the Base Case and smart charging scenarios because of the significant increase in demand during peak periods (Fig. 2). The plan outcome of ADEQ-OPTSELECT comprises one substation upgrade, five DG units, six capacitors, 11 feeder upgrades and the optimal DR; and one iteration is needed to arrive at the final plan. It is noted that DR is allocated in load blocks 8 and 9 which have the highest LSF and only at the remote buses. The outcome of ADEQ-OPTPERIOD presented in Table 9 shows that there is a need for DR capacity starting from year 7 until the plan terminal year because of the increasing penetration of PEVs.

5.3. Summary of 32-bus system studies

A comparison of the distribution system plan outcomes obtained from the Base Case, smart PEV charging and uncontrolled PEV charging cases are presented in Table 10. It is noted that the present worth of the plan cost in the PEV smart charging case is slightly higher than the Base Case, while much higher for uncontrolled charging. The results also reveal that the impact of PEV penetration are much damped in terms of added cost and added capacity when smart charging is considered as compared

Table 10

LDC plan comparison for 32-bus case studies.

	Case-1	Case-2a	Case-2b
Present worth of total cost (M\$)	8.64	9.59	14.92
Added capacity (MVA)	24.7	30.7	38.8
Reserve margin, γ	0.21	0.31	0.54

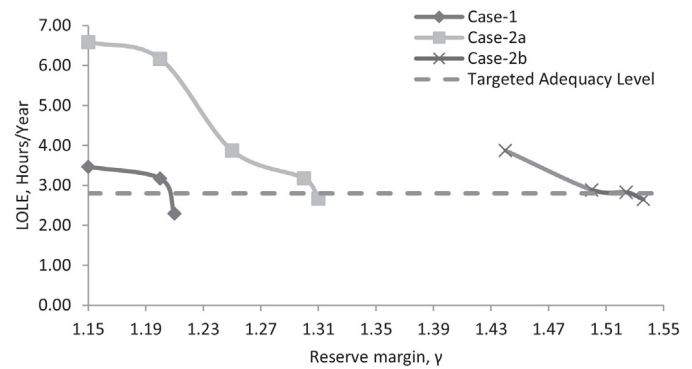


Fig. 3. LOLE for different reserve margins, with and without PEVs.

to uncontrolled charging. The proposed model can be utilized by LDCs to determine the allowable PEV penetration limits for a given distribution system infrastructure, expected impacts on system operation, the capital spending, and the required upgrades in the distribution network in case of PEV smart charging (best case scenario) and uncontrolled charging (worst case scenario).

In Fig. 3, the variation of LOLE with γ and the optimal value of γ , for the three considered case studies are presented. It is noted that the optimal value of γ is reached when γ is below the dashed line, the Targeted Adequacy Level. It is noted that γ increases from 0.21 in the Base Case to 0.54 with uncontrolled PEV charging, indicating a high reserve requirement because of the significant increase in charging demand during peak periods due to high PEV arrival rates. However, when PEV smart charging is considered, optimal γ is 0.31 and the burden on the system planner is much lower with reduced peak demand compared to Case-2b (uncontrolled charging) because the charging demand is mostly scheduled to off-peak periods.

5.4. Case-3: Base Case; 69 bus test system

It is to be noted that only Base Case studies are carried out for this test system. The results reveals that the targeted adequacy level is met with $\gamma = 0.17$; the selected upgrades from ADEQ-OPTSELECT comprises one substation, two DG units, two capacitors and two feeders. Two iterations are needed to arrive to the optimal plan for the terminal year. The final plan, after executing ADEQ-OPTPERIOD, are presented in Table 11.

5.5. Computational aspects

The Smart-DSPLAN is an MINLP model which is solved using the COINBONMIN solver which uses the branch and bound, branch and cut, and outer approximation algorithms. The Smart-DSPLAN2 is an NLP model solved using the SNOPT solver which uses a sequential quadratic programming algorithm that obtains search directions

Table 11

Case-3 results of ADEQ-OPTPERIOD.

Year	Upgrades size (MW) and site (Bus)
10	–
9	–
8	(0.5) Fdr 1–2 and (0.5) Fdr 2–3
7	–
6	(0.4) DG #64
5	–
4	–
3	–
2	(0.6) DG #61
1	Capacitors: (0.1) #61 and (0.2) #62, and (1.3) Substation

Table 12
Model statistics.

Case study	Model	Single variables	Discrete variables	CPU time, per iteration, s	Total CPU time, s
Case-1	Smart-DSPLAN	1840	98	770	10,783
	Smart-DSPLAN2	2535	0	0.22	5
Case-2a	Smart-DSPLAN	2071	98	721	11,534
	Smart-DSPLAN2	2535	0	0.25	4
Case-2b	Smart-DSPLAN	2597	98	496	6950
	Smart-DSPLAN2	3259	0	1.75	35
Case-3	Smart-DSPLAN	3800	206	1061	6363
	Smart-DSPLAN2	5258	0	11	284

from a sequence of quadratic programming sub-problems. The considered test system is programmed and executed on a Dell PowerEdge R810 server, in GAMS environment, Windows 64-bit operating system, with 4 Intel-Xeon 1.87 GHz processors and 64 GB of RAM.

The model and solver statistics are given in Table 12. It can be seen that Smart-DSPLAN requires more CPU time per iteration as compared to Smart-DSPLAN2 in all case studies, because of the presence of binary variables in Smart-DSPLAN. In addition, the 69-bus test system requires more time per iteration compared to the 32-bus test system due to the increased number of buses and hence the number of variables and constraints. Moreover, the total CPU time varies across the case studies as it depends on the number of iterations to arrive at the optimal solution, and the required CPU time per iteration.

6. Conclusions

In this work, a comprehensive framework for multi-year distribution system planning was presented. The proposed framework aimed to determine the optimal upgrade plan, siting and sizing of DGs, substations, capacitors and feeders in distribution systems, considering the penetration of PEVs in an uncontrolled as well as smart charging environment, and DR options. Two criteria, BCR and adequacy analysis, were incorporated within the proposed framework, to determine the optimal distribution system plan. The uncontrolled PEV charging load model considered driver behaviour, arrival time, mileage, and other information to develop a charging load profile for inclusion in the planning problem. Similarly, a novel set of PEV operational constraints were included, in the planning model, to capture PEV smart charging effects. The proposed planning framework was applied to two test systems. Four case studies were considered to investigate the impact of PEV smart and uncontrolled charging loads as well as DR options on the distribution plan.

The studies reveal that it is important for planners to take into consideration the effects of PEV penetration while deriving their plan outcomes, particularly, when the share of uncontrolled PEV charging is high. It was noted that the present worth of the plan cost in the PEV uncontrolled charging case was much higher than that in the Base Case (No PEV), while the impact of PEV penetration was much damped in terms of added cost and added capacity when smart charging was considered. Compared to the Base Case, with uncontrolled PEV charging loads, there was a 77% and 63% increase in the present worth of the plan cost and in the added capacity (in MW), respectively. The reason behind this significant increase is that the system peak demand coincided with the uncontrolled charging demand due to the coincident home arrival rates during peak hours. On the other hand, it was noted that there was a 11% and 24% increase in the present worth of plan cost and in the added capacity (in MW) when smart charging was used. There was a decrease in peak demand compared to uncontrolled charging

because of smart charging which scheduled the charging demand to off-peak periods. The proposed framework can be used by LDCs to quantify the impacts, and the required upgrades in the distribution networks; and be readily applied to any radial distribution system. It is also noted that although the present work considered distribution systems connected to the grid, the proposed framework is easily extendable to isolated/islanded distribution systems or microgrids, without any loss of generality.

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